Unit: 6 Unit 4: Absolute Value
Topic: Solving Absolute Value Equations

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. Solve the equation $|x| = 5$.
(A) $x = 5$
(B) $x = -5$
(C) $x = 5$ or $x = -5$
(D) $x = 0$
(E) $x = 10$

Q2. Solve the equation $|2x - 3| = 7$.
(A) $x = 2$ or $x = 5$
(B) $x = -2$ or $x = -5$
(C) $x = 2$ or $x = -5$
(D) $x = 5$ or $x = -2$
(E) $x = 0$

Q3. Solve the equation $|x + 2| = 4$.
(A) $x = 2$
(B) $x = -2$
(C) $x = 2$ or $x = -6$
(D) $x = -2$ or $x = 6$
(E) $x = 0$

Q4. Solve the equation $|3x| = 12$.
(A) $x = 4$
(B) $x = -4$
(C) $x = 4$ or $x = -4$
(D) $x = 0$
(E) $x = 12$

Q5. Solve the equation $|x - 1| = 3$.
(A) $x = 2$ or $x = 4$
(B) $x = -2$ or $x = -4$
(C) $x = 2$ or $x = -2$
(D) $x = 4$ or $x = -4$
(E) $x = 1$

Q6. Solve the equation $|2x + 1| = 9$.
(A) $x = 4$ or $x = -5$
(B) $x = -4$ or $x = 5$
(C) $x = 4$ or $x = -4$
(D) $x = -4$ or $x = -5$
(E) $x = 0$

Q7. Solve the equation $|x| = |2x - 3|$.
(A) $x = 1$ or $x = 3$
(B) $x = 1$ or $x = -1$
(C) $x = 3$ or $x = -3$
(D) $x = 0$
(E) $x = 2$

Q8. Solve the equation $|x + 4| = |x - 2|$.
(A) $x = 1$
(B) $x = -1$
(C) $x = -3$
(D) $x = 3$
(E) $x = 0$

Open Ended Questions (FRQ)

Q9. Solve the equation $|2x - 5| = |x + 1|$ and check for extraneous solutions.

Q10. Solve the equation $|x + 2| = |3x - 4|$ and check for extraneous solutions.

Chef's Tips

- Check for extraneous solutions
- Use the definition of absolute value
- Consider all possible cases